

Cancelling Translated Copies of Points in Convex Position

Problem Statement

Definition 1. A finite set S of points in the plane is in convex position if every point of S is a vertex of its convex hull.

Definition 2 (Setup). Let S be a finite set of $n \geq 1$ points in $\mathbb{R}^2$ in convex position. Let T be a finite multiset of translating vectors in $\mathbb{R}^2$, each assigned a sign $\sigma(t) \in \{+1, -1\}$.

For every point $p \in \mathbb{R}^2$, define

$$\sigma(p) = \sum_{\substack{t \in T \\ \exists s \in S: p=t+s}} \sigma(t).$$

Theorem 1 (Lower Bound – Known). Under the above setup, there are at least n points $p \in \mathbb{R}^2$ for which $\sigma(p) \neq 0$.

Conjecture 1 (Main – To Be Proved). If there are exactly n points $p \in \mathbb{R}^2$ for which $\sigma(p) \neq 0$, then S is the set of vertices of a parallelogram (in particular, $n = 4$).

Goal

Prove the conjecture from scratch. That is, assuming the lower bound holds (or re-proving it as part of the argument), show that equality $|\{p : \sigma(p) \neq 0\}| = n$ forces S to be a parallelogram (and hence $n = 4$).

Remark 1. The lower bound follows from convex geometry: the vertices of the convex hull of $S + T := \{t + s : t \in T, s \in S\}$ (counted with signed multiplicity) cannot all cancel. The content of the conjecture is that the extremal case forces S to be a parallelogram.